

Frozen Hub Caption Transfer — corrected and expanded

How a caption-retrieval head trained only from DINOv2-small is evaluated on ConvNeXt. Checked against the Final Project Report (69 pp) and the cached artifacts (hub_exact.npz, G10 output). Corrections to the source document are marked; new sections answer questions the source left open.

Core claim, unchanged and correct. ConvNeXt never passes through DINOv2-small at test time. It gets its own fitted entry map into the frozen hub; the frozen caption head was trained using DINOv2-small hub coordinates only. The 96.7 percent is $R@1(\text{transferred}) / R@1(\text{native ConvNeXt head})$ — a retention ratio, not a raw $R@1$.

Part A — corrections to the source document

where	the source says	correction, with evidence
Models table: 'original hub sources'	DINOv2-small, -base, -large, BGE-M3 and GPT-2 (five spaces); seven-space variant adds BERT and SBERT	Three hub configurations exist, and the source conflates them. The 96.7 percent ConvNeXt figure (E.2) was measured on the FOUR-space hub: DINOv2 s/b/l + bge-m3, concat width 5,376. Evidence: hub_exact.npz has space_names of length 4 and mu of length 5,376 = 768+1536+2048+1024 — no GPT-2. The five-space hub (6,144-d, adds GPT-2) is Section C.13's body; the seven-space hub (7,680-d, adds BERT and SBERT) is C.13.3 and G1. ConvNeXt reaches 96.7 percent on the four-space hub and 99.3 percent on the seven-space one (G10).
'Protocol caveat' on the split	If PCA/whitening or entry-map fitting used all 9,533 rows, the protocol is transductive and should be disclosed	Resolved — it did not. hub_exact.npz carries hub_train of shape (8533, 512), train_idx (8533) and eval_idx (1000) as separate arrays. The PCA basis, the whitening, every entry map and the head are fitted on the 8,533 training rows; the 1,000 held-out rows are transformed and scored only. The report states this at C.13: '1,000 rows held out for all measurements, never used in fitting'. The caveat can be closed, not disclosed.
Stage 2, 'normalize per encoder'	$Z_k = (X_k - \mu_k) / \sigma_k$, described as normalising 'that encoder's columns'	Half right. The mean μ is per column (length 5,376). The scale is one scalar per space — scale_per_space has shape (4,), not (5376,). So this is per-space rescaling, not per-column z-scoring. The distinction matters because per-column whitening would happen anyway in step 4; the per-space scalar is what stops one encoder's overall magnitude dominating the PCA. See Part C for why that is load-bearing.
Retrieval section	Cosine similarity ranks 1,000 gallery captions; $R@1$ counts top-rank hits	Correct. Two additions worth having: chance is $1/1000 = 0.001$, and the random-entry control lands at 0.001 — exactly chance — so the entire 96.7 percent is carried by the fitted entry map and the hub.
'Why this is notable'	Different architecture, dimension and objective; above the 93.8–96.5 within-family band	Correct, and there is a sharper point since measured (E.12): ConvNeXt's raw shape agreement with every other encoder is the lowest in the roster (0.30–0.40 against text, 0.39 against the DINOv2 family, versus 0.56–0.60 among DINOv2 siblings). It transfers best while agreeing least. The hub recovers a correspondence the raw geometry does not show. See Part D.
bge-m3 listed as both hub source and caption target	Stated without comment	This dual role is a measured dependency, not an oversight: the head's target sitting inside the hub is worth +3.0 percentage points beyond what any text member contributes (G10 Q2, declared in the report at C.13.2 and E.6). Held-out scoring rules out leakage. Worth stating so it cannot be found first by a reader.

Part B — what '512' means, and what it does not

A natural misreading of 'the head is trained on DINOv2-small hub vectors' is that some *selection* of vectors takes place — the 512 items whose DINOv2-small vectors best match their captions by cosine, say, or the 512 most representative rows. **Nothing of the kind happens.** The number 512 is a *width*, not a *count*:

quantity	value	what it is
512	columns	the number of principal components kept after PCA on the concatenated space. Every item's hub coordinate is a vector of 512 numbers.
8,533	rows	the number of items whose DINOv2-small hub coordinates train the head. All of them. No filtering, no scoring, no cosine threshold.
1,024	columns	the width of the target: each item's bge-m3 caption vector.
W_{head}	$512 \times 1,024$	the single matrix fitted by ridge regression so that $H_{\text{small}} W_{\text{head}} \approx T_{\text{bge}}$ over all 8,533 rows at once.

The head is fitted on the full training set by one closed-form solve: $W_{\text{head}} = (H_{\text{small}}^T H_{\text{small}} + \alpha I)^{-1} H_{\text{small}}^T T_{\text{bge}}$, with H_{small} of shape $8,533 \times 512$ and T_{bge} of shape $8,533 \times 1,024$. Every training row contributes to the least-squares objective with equal weight. There is no notion of 'best' rows — a well-matched cat/caption pair and a poorly matched one both enter the sum, and ridge finds the single matrix that minimises total squared error across all of them plus the $\alpha||W||^2$ ridge penalty.

Two things that *do* select, and could be confused with selecting rows:

mechanism	what it selects	what it does not select
PCA to 512 components	directions in the concatenated space: the 512 orthogonal axes carrying the most variance across all 8,533 training items	any subset of items. Every item is projected onto all 512 axes.
The head's ridge fit	nothing — it uses every row	rows by quality, cosine, or any score
Retrieval at test time	the top-ranked caption for each of the 1,000 held-out queries, by cosine	anything during training; cosine ranking is an evaluation step only

So the sentence 'the head is trained on DINOv2-small's hub coordinates' means: take all 8,533 training images, run DINOv2-small once, multiply by W_{small} to get 8,533 vectors of width 512, and fit one $512 \times 1,024$ matrix that maps those to the corresponding 8,533 bge-m3 caption vectors. 'Cosine' appears nowhere in training.

Part C — two design choices the source states without explaining

Why 512 and not 768, or 1,024? Not because 512 is optimal — it is not tuned. It is chosen to sit safely *below* the algebraic rank of the narrowest source encoder. DINOv2-small is 768-d, so its hub coordinates have rank at most 768. At hub width 768 exactly, the last retained singular value is near-null ($\sigma = 0.0054$ against 10.79 one step earlier), and the least-squares head divides by it: transfer holds at 0.894 through $k = 760$ and collapses to 0.009 at $k = 768$. That is the width cliff of report E.8, resolved to a single direction and confirmed on DINOv2-base out of sample. The rule every hub inherits is $\text{hub_width} < \text{dim}(\text{source})$; 512 satisfies it with margin. A truncated or more heavily regularised head lifts the constraint entirely, so 512 is a safe operating point, not a ceiling.

Why per-space scaling is load-bearing. The raw encoder outputs differ in magnitude by more than two orders. Mean row norms, measured in G10:

encoder	mean row norm	consequence without per-space scaling
GPT-2	208.21	would dominate the PCA almost entirely — 234× the smallest
DINOv2-small / -base / -large	54.70 / 52.91 / 50.83	the image side, roughly comparable to one another
BERT	9.19	
SBERT	2.43	
bge-m3	0.89	would be nearly invisible to the PCA — yet it is the head's target

Concatenating these raw would let GPT-2's 768 columns outweigh bge-m3's 1,024 columns by roughly 234× in the variance PCA sees. One scalar per space equalises the overall magnitudes first; the per-column whitening in step 4 then equalises the retained directions. Both steps are fitted on the 8,533 training rows only.

Part D — the pipeline, end to end, with the corrections folded in

step	math	fitted on	what it produces / notes
1. cache	X_k for k in {DINOv2-s, -b, -l, bge-m3}	all 9,533 rows run forward once (GPU)	raw vectors; the only GPU step in the project
2. centre + scale	$Z_k = (X_k - \mu_k) / s_k$	μ per column, s one scalar per space, both from 8,533 train rows	four spaces of comparable magnitude
3. concatenate	$Z = [Z_s \mid Z_b \mid Z_l \mid Z_{bge}]$, 5,376-d	—	one row per item
4. PCA + whiten	$H = Z V_{512} \Sigma_{512}^{-1}$	SVD of the 8,533 train rows; V and Σ frozen	THE HUB: an $8,533 \times 512$ whitened basis, extended to the 1,000 test rows by the same V, Σ
5. entry maps	$W_k = \text{ridge}(X_{k,\text{train}}, H_{\text{train}})$, $\alpha = 1e-2$	train rows	one per encoder; W_{small} is 768×512
6. head	$W_{\text{head}} = \text{ridge}(X_{\text{small},\text{train}}, W_{\text{small}}^T T_{\text{bge},\text{train}})$	all 8,533 train rows of DINOv2-small only	$512 \times 1,024$; frozen from here on
7. ConvNeXt arrives	$X_{\text{conv}} (9,533 \times 1,024)$ cached; $W_{\text{conv}} = \text{ridge}(X_{\text{conv},\text{train}}, H_{\text{train}})$	train rows	the ONLY thing fitted for ConvNeXt. Hub, head, other maps untouched
8. transfer	$T\text{-hat}_{\text{transfer}} = X_{\text{conv},\text{test}} W_{\text{conv}} W_{\text{head}}$	nothing — evaluation	predicted bge-m3 vectors for the 1,000 held-out images
9. native reference	$W_{\text{native}} = \text{ridge}(X_{\text{conv},\text{train}}, W_{\text{conv}}^T T_{\text{bge},\text{train}})$; $T\text{-hat}_{\text{native}} = X_{\text{conv},\text{test}} W_{\text{conv}} W_{\text{native}}$	train rows	the best a linear head can do with full access to ConvNeXt's own hub coordinates
10. score	$R@1$ of each against the 1,000-caption bge-m3 gallery by cosine; report $R@1_{\text{transfer}} / R@1_{\text{native}}$	—	96.7 percent (four-space hub); 99.3 percent on the seven-space hub. Random W_{conv} : 0.001 = chance

What is zero-shot, precisely. The head (step 6) is zero-shot with respect to ConvNeXt: it was fitted before ConvNeXt existed in the project and is applied unchanged. The entry map (step 7) is *not* zero-shot — it is one closed-form solve on ConvNeXt's training-row vectors. The claim is about the head. A reader who hears '96.7 percent zero-shot' should understand: one matrix was fitted for ConvNeXt, and everything downstream of it was reused.

Part E — the finding the source could not have known: raw agreement does not predict transfer (E.12)

Before any hub or map exists, one can ask how well two encoders' raw geometries agree: the Spearman correlation between their pairwise-distance structures over all 9,533 items, with nothing fitted. Measured across eight encoders (the seven hub members plus ConvNeXt, 28 pairs):

ConvNeXt-base against	raw ρ	for comparison
DINOv2-small / -base / -large	0.397 / 0.392 / 0.391	DINOv2 variants agree with each other at 0.56–0.60
SBERT (its best text pair)	0.330	DINOv2-large's best text pair is BERT at 0.418
BERT / bge-m3	0.305 / 0.302	
GPT-2	0.166	the collapsed space; lowest for every encoder

ConvNeXt has the **lowest** raw agreement of any image encoder — below DINOv2-large against text, and even its agreement with the other image encoders (0.39) is well under how the DINOv2 variants agree among themselves. Its provenance predicts this: a convnet trained on 22,000 human labels, differing from everything else on architecture and supervision at once. And it is the best-transferring encoder measured. The 96.7 percent is therefore not underwritten by superior raw geometric agreement; it is achieved despite the weakest raw agreement in the set. The hub and its entry map recover a linear correspondence that the raw shape-agreement metric does not register.

What this does and does not establish. One encoder is a demonstration, not a law. Shape agreement fits nothing; transfer fits an entry map on training rows — so this compares what raw geometry reveals with what a fitted map achieves, not two estimates of one quantity. It removes an assumption that was never tested, that hub transfer tracks raw agreement, and records that it does not. It is the second instance of this lesson: E.6 found the same on the text side, where uniformly low reach (R^2 0.02–0.05) failed to predict an 11.2-point spread in transfer.

Part F — the one-page mental model, corrected

claim	precise version
'The head never trained on ConvNeXt.'	True. W_{head} is fitted from H_{small} to T_{bge} over all 8,533 training rows and never sees a ConvNeXt vector.
'The head trained on the 512 best DINOv2-small vectors.'	False. 512 is the hub width. All 8,533 training rows train the head; nothing is selected by cosine or any score.
'ConvNeXt never trained in any form.'	Too broad. W_{conv} is fitted from X_{conv} to H on training rows. Only the head is untouched.
'ConvNeXt goes through DINOv2-small.'	False. ConvNeXt $\rightarrow W_{\text{conv}} \rightarrow \text{hub} \rightarrow W_{\text{head}}$. DINOv2-small is not on the runtime path; its role was to supply the head's training inputs.
'96.7 percent transfer.'	R@1 retention against a ConvNeXt-specific native linear head, on the four-space hub. 99.3 percent on the seven-space hub.
'All 9,533 were training rows.'	False, and confirmed from the cache: 8,533 fit everything; 1,000 are held out and only ever transformed and scored.
'Why 512?'	Below the narrowest source's rank (768) with margin; at 768 exactly the fit divides by a near-null singular value (E.8). Not tuned, not optimal, safe.
'ConvNeXt transfers well because its geometry already agrees.'	False. Its raw agreement is the lowest in the roster (E.12). The hub is recovering structure the raw metric does not show.

Suggested project statement. After ConvNeXt is independently aligned to the frozen four-space hub with one closed-form entry map, a caption-retrieval head fitted only on DINOv2-small's hub coordinates — all 8,533 training rows, no selection — retains 96.7 percent of the R@1 of a ConvNeXt-specific linear head. Every learned transform is fitted on the 8,533 training rows; retrieval is scored on 1,000 held-out rows against a 1,000-caption gallery at chance 0.001. ConvNeXt's raw geometry agrees least with the other encoders and transfers best.